Worksheet: same evolution, different probabilities

Supplementary material for the paper by A. Yu. Polushkin. One tutorial; a calculator is enough.

Setting. On the circle $0 \le \theta < 1$ take $\psi(\theta) = c_0 + c_1 e^{2\pi i\theta}$ with $|c_0|^2 + |c_1|^2 = 1$. A detector reports outcome 0 for $\theta \in [0, 1/2)$ and outcome 1 for $\theta \in [1/2, 1)$. On a grid with nodes $\theta_j = (j + \xi)/Q$ the node probabilities are $|\psi(\theta_j)|^2/Q$, and $\widetilde{p}_0$ is their sum over nodes with $\theta_j < 1/2$. A node at $1/2$ belongs to outcome 1. Translation by a/Q permutes the nodes and multiplies c_1 by $e^{2\pi i a/Q}$. The rotation to be approximated has $\rho = (\sqrt{5} - 1)/2$.

A. Predict before calculating.

Angle and grid (midpoint nodes, $\xi = 1/2$)	$\lvert a/Q - \rho\rvert$
$3/5$ on $Q = 5$	0.018034
$6/10$ on $Q = 10$	0.018034
$4/6$ on $Q = 6$	0.048633

A1. Must the first two rows predict the same detector probabilities? Which grid predicts them better, the first or the third, after one step? Write down your answers.

A2. Besides the angle error, what must be specified before the question has an answer?

B. The exact detector.

B1. Find the 2×2 matrix E with $p_0 = c^* E c$, and its eigenvalues.

B2. Show that $E^2 \ne E$. Why does this not contradict a projective coordinate measurement on the full circle?

B3. Find p_0 for $c = (1, e^{i\phi})/\sqrt{2}$.

C. Node sums by hand.

C1. How many midpoint nodes lie in $[0, 1/2)$ for odd and for even Q? What is $\widetilde{p}_0$ for the constant wavefunction?

C2. Take $c = (1, -1)/\sqrt{2}$, so that $|\psi|^2 = 1 - \cos 2\pi\theta$. Compute $\widetilde{p}_0$ for midpoint grids with $Q = 5$ and $Q = 10$. Compare with the exact value.

C3. Take $c = (1, 1)/\sqrt{2}$ and $Q = 8$. Compute $\widetilde{p}_0$ for nodes $j/8$ and for midpoint nodes. Both grids implement the $5/8$ translation as a permutation. What differs?

D. The worst input. Write the grid effect as $\begin{pmatrix} 1/2 + d & g \\ g^* & 1/2 + d \end{pmatrix}$.

D1. Use a geometric sum to find d and g for midpoint grids, separately for odd and even Q.

D2. With $\delta = a/Q - \rho$, the error matrix after k steps, in the frame with the ideal evolution removed, has diagonal d and off-diagonal $h_k = g\, e^{2\pi i k\delta} - i/\pi$. Show that the largest error over all inputs is $\varepsilon_k = |d| + |h_k|$, and construct an input that attains it.

D3. Calculate ε_0 and ε_1 for the three rows above. Which of your answers in part A survive? How does this worst-case statement differ from your results in C2?

E. Repair and explain.

E1. Show that the nodes with $\xi = 1/4$ and $\xi = 3/4$ together form the midpoint grid with $2Q$ nodes, and hence $\frac{1}{2}(E^{Q,1/4} + E^{Q,3/4}) = E^{2Q,1/2}$.

E2. Calculate ε_0 and ε_1 for $3/5$ on $Q = 5$ when ξ is chosen at random between these two values. Which quantity has not changed?

E3. For $Q = 5$, compute d for each setting alone. Is the improvement a property of each run? Does the random choice reduce the number of runs needed?

F. In your own words. State what was held fixed in each comparison. Why is "avoid a boundary node" an incomplete summary of the lesson?